



Data Converters

Chapter 02

Static Metrics & Nonidealities

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Dr.-Ing. Matthias Keller

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Laboratory for Microelectronics
Department of Microsystems Engineering – IMTEK
University of Freiburg



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VLSI Data Conversion Circuits

held by

Prof. B. Murmann, PhD, University of Hawai'i

during his sabbatical at the University of Freiburg in 2009



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Static Nonidealities

Static deviations of transfer characteristics from ideality:

- offset
- gain error
- differential nonlinearity (DNL)
- integral nonlinearity (INL)

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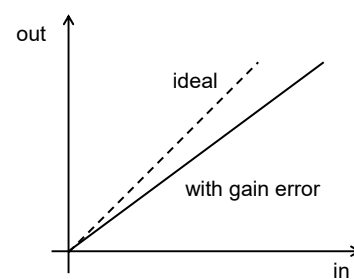
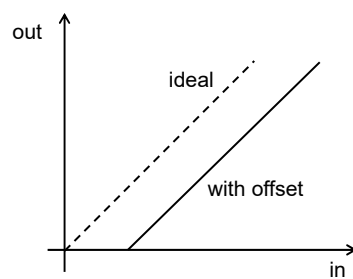
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Offset & Gain Error

general idea (neglecting staircase nature of transfer functions)



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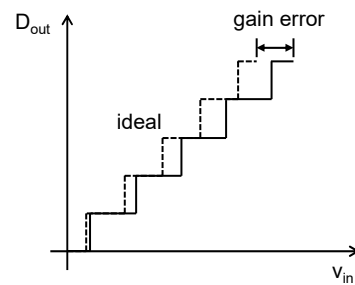
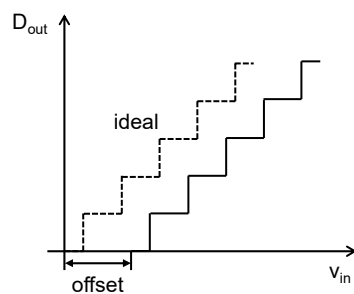
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Offset & Gain Error – ADC

definitions based on endpoints of transfer characteristic

- $\frac{1}{2}$ LSB before (after) first (last) transition
- offset is the deviation of bottom endpoint from its ideal location
- gain error is the deviation of the top endpoint from its ideal location with offset removed



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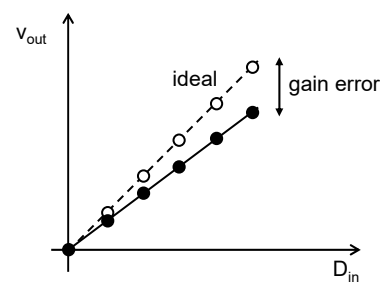
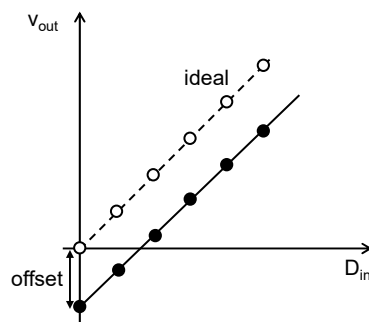
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Offset & Gain Error – DAC

- same idea, endpoints are directly defined by analog output values at minimum and maximum digital input
- errors are specified along the vertical axes



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Differential Nonlinearity – DNL

- In an ideal world, all ADC codes would have equal width; all DAC output increments would have same size.
- $DNL(k)$ is a vector that quantifies for each code k the deviation of this width from the “average” width (step size).
- $DNL(k)$ is a measure of uniformity, it does not depend on gain and offset errors
→ scaling and shifting a transfer characteristic does not alter its uniformity and thus $DNL(k)$.

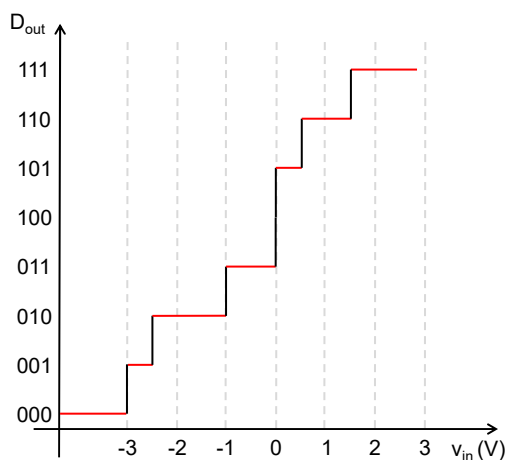
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Example DNL – 3-bit ADC



code	width W (V)
111	undefined
110	1
101	0.5
100	0
011	1
010	1.5
001	0.5
000	undefined

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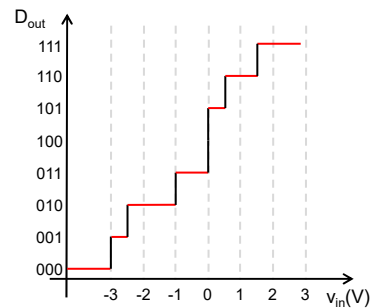
Example DNL – 3-bit ADC

- average code width
 - ADC with perfect uniformity would divide range between first and last transition into 6 equal steps
 - average code width (= LSB size) results in

$$W_{avg} = \frac{1.5V - (-3V)}{6} = 0.75V$$

- calculate DNL(k) for each code based on

$$DNL(k) = \frac{W(k) - W_{avg}}{W_{avg}}$$



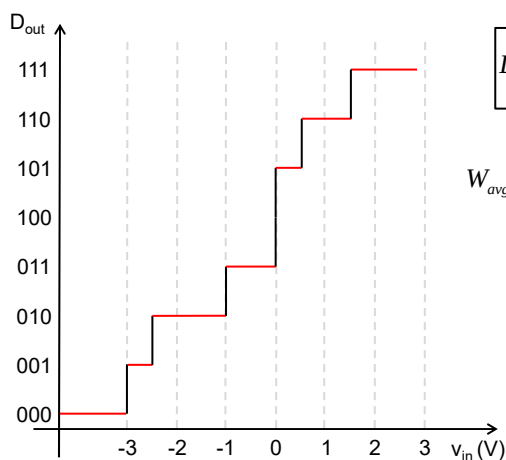
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Example DNL – 3-bit ADC



$$DNL(k) = \frac{W(k) - W_{avg}}{W_{avg}}$$

$$W_{avg} = \frac{1.5V - (-3V)}{6} = 0.75V$$

code	width W (V)	DNL (LSB)
111	undef.	undef.
110	1	0.33
101	0.5	-0.33
100	0	-1
011	1	0.33
010	1.5	1
001	0.5	-0.33
000	undef.	undef.

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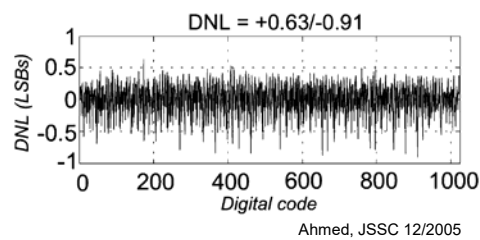
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Example DNL – 3-bit ADC

- positive (negative) DNL implies wide (narrow) code
- DNL = -1 LSB implies missing code
- impossible to have DNL < -1 LSB for an ADC
- DNL typically given in terms of min/max number across all codes



code	DNL (LSB)
111	undefined
110	0.33
101	-0.33
100	-1
011	0.33
010	1
001	-0.33
000	undefined

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DNL – DAC

same approach

- calculate output increments for each digital code
- calculate increment that divides range into equal steps
- calculate DNL for each code k based on

$$DNL(k) = \frac{step(k) - step_{avg}}{step_{avg}}$$

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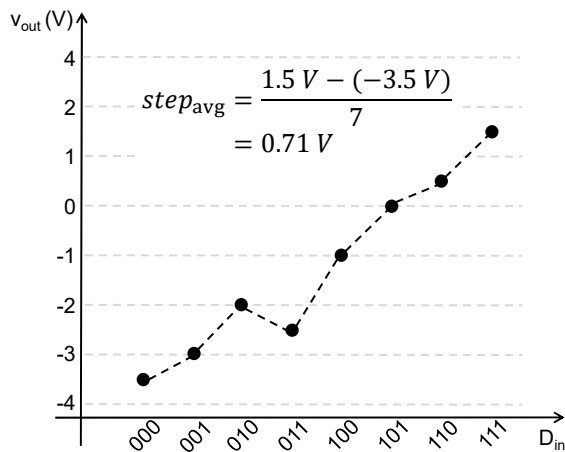
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Example DNL – 3-bit DAC



step (k)	DNL (LSB)
7	0.41
6	-0.30
5	0.41
4	1.11
3	-1.7
2	0.41
1	-0.30

DNL < -1 LSB
→ non-monotonic DAC

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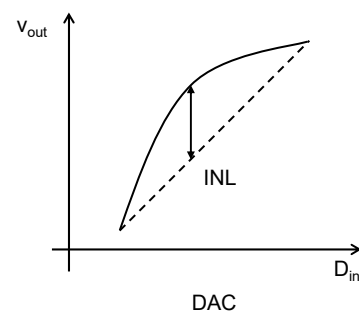
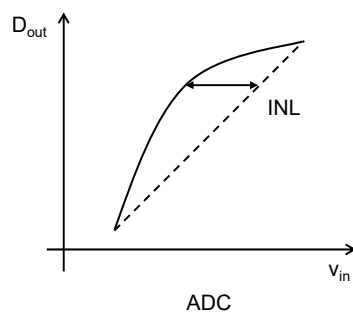
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Integral Nonlinearity – INL

For each “relevant point” of the transfer characteristic, quantify distance from a straight line drawn through the endpoints.

As with DNL, INL is independent of gain and offset errors.



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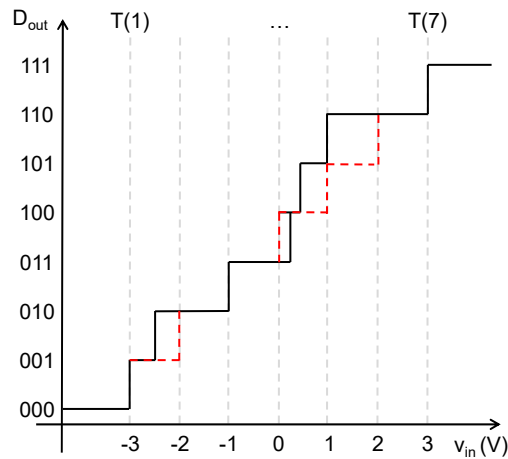
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Example INL – 3-bit ADC



- “straight line” reference is uniform staircase between first and last transition
- INL for each code is

$$INL(k) = \frac{T(k) - T_{uniform}(k)}{W_{avg}}$$

- INL(0) is undefined
- obviously: $INL(001) = INL(111) = 0$

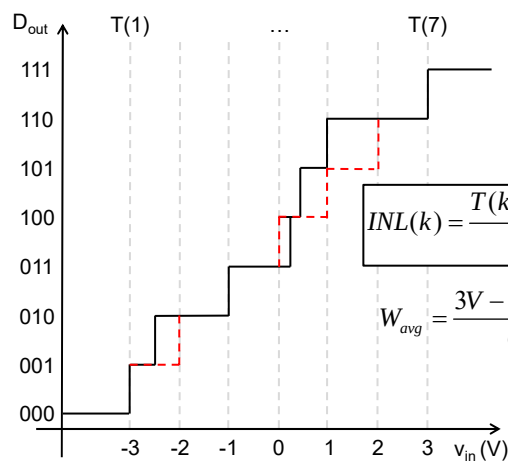
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Example INL – 3-bit ADC



code	T(k)	T _{uniform} (k)	INL (LSB)
111	3	3	0
110	1	2	-1
101	0.5	1	-0.5
100	0.25	0	0.25
011	-1	-1	0
010	-2.5	-2	-0.5
001	-3	-3	0
000	undef.	undef.	undef.

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Example INL – 3-bit ADC

Note that $INL(k) = \sum_{i=1}^{k-1} DNL(i)$

$$DNL(k) = \frac{W(k) - W_{avg}}{W_{avg}}$$

$$INL(k) = \frac{T(k) - T_{uniform}(k)}{W_{avg}}$$

$$W_{avg} = \frac{3V - (-3V)}{6} = 1V$$

code	DNL (LSB)	INL (LSB)
111	undefined	0
110	1	-1
101	-0.5	-0.5
100	-0.75	0.25
011	0.25	0
010	0.5	-0.5
001	-0.5	0
000	undefined	undefined

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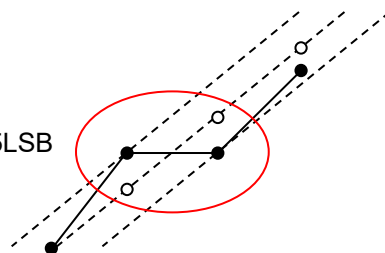
INL – DAC

- same idea
 - find ideal output values located on a straight line between endpoints
 - calculate INL for each code

$$INL(k) = \frac{V_{out}(k) - V_{out,uniform}(k)}{step_{avg}}$$

- if for all codes $|INL| < 0.5 \text{ LSB}$
 $\rightarrow DNL \geq -1 \text{ LSB} \rightarrow \text{monotonicity}$

worst case: $+0.5\text{LSB} \rightarrow -0.5\text{LSB}$
 $\rightarrow \text{flat}$



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